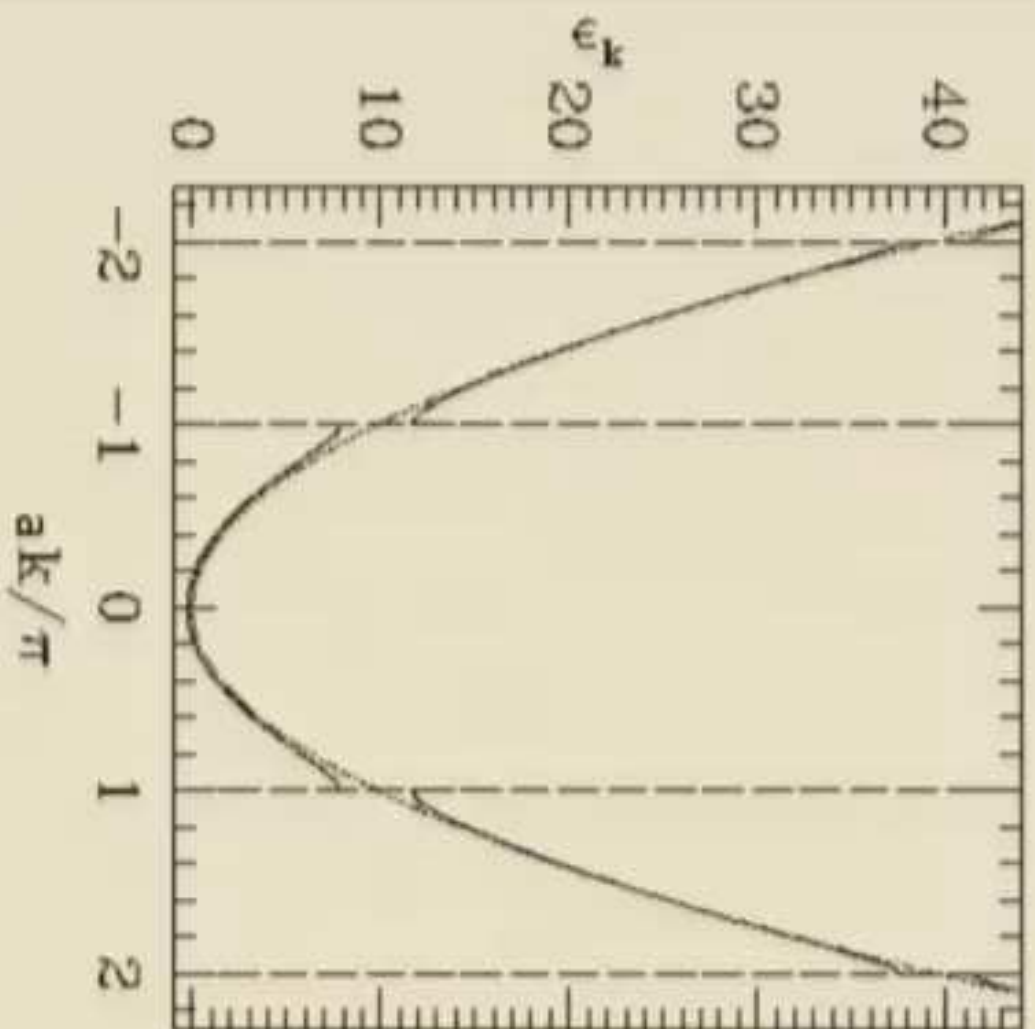
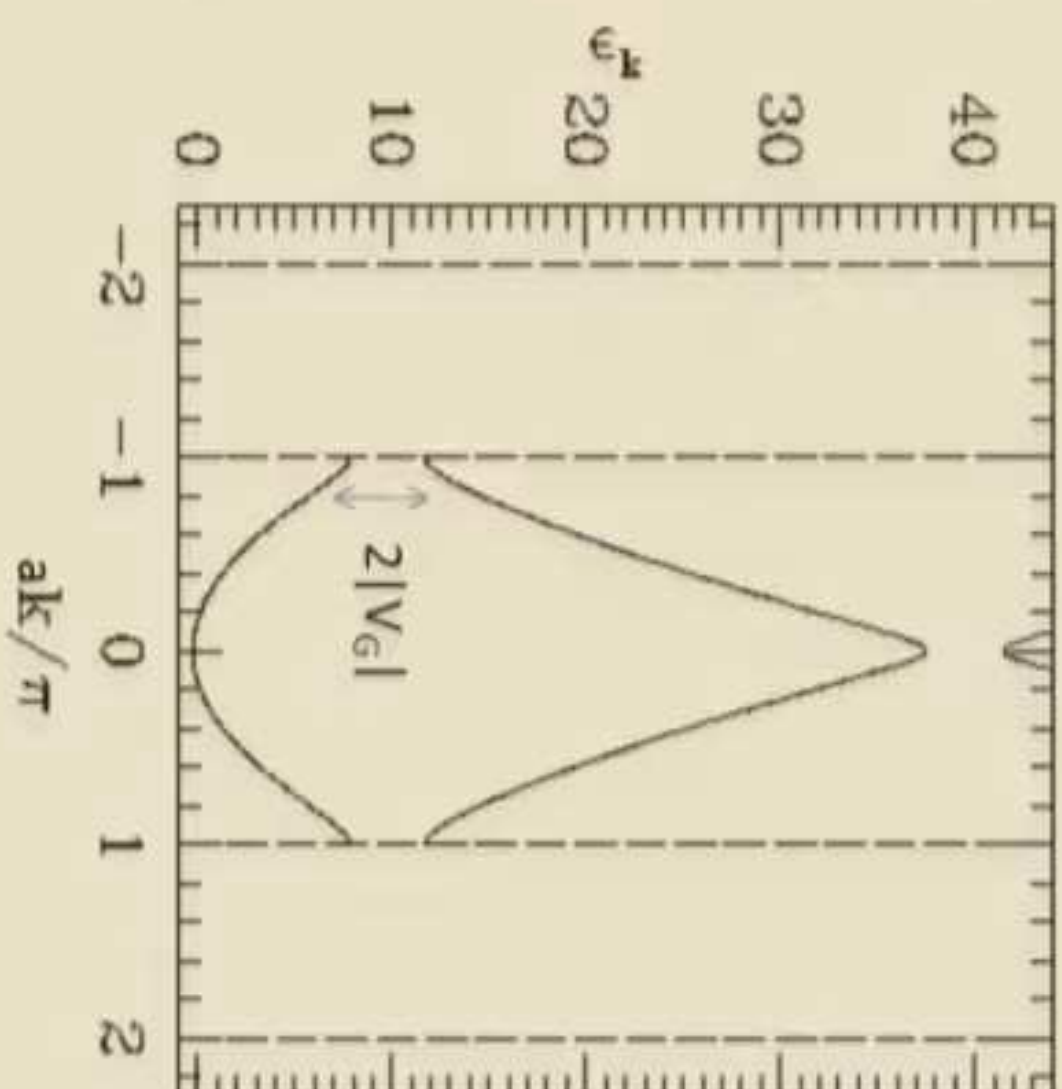




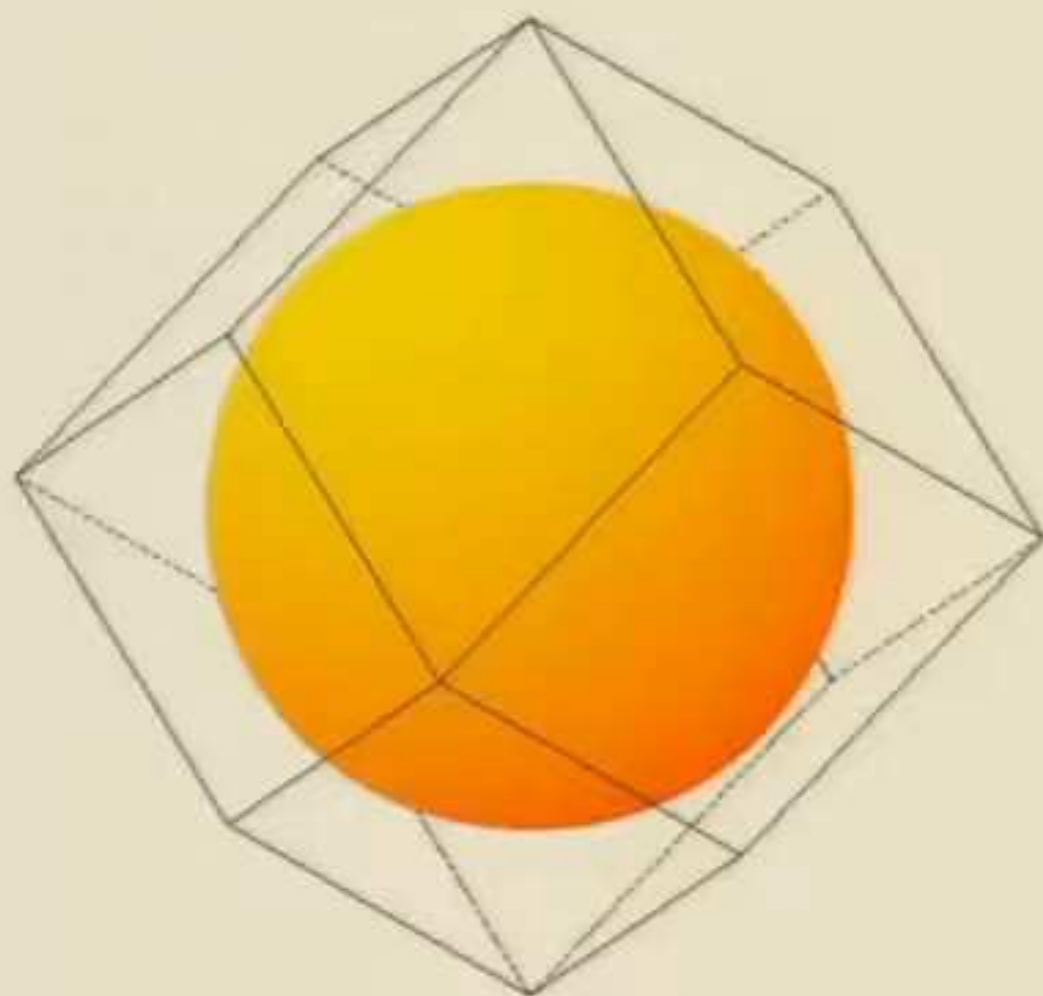
$d=1$ extended zone



$d=1$ reduced zone



Na
bcc $a=7.96\text{\AA}$
1st band
electron Fermi surface
Fermi surface
file name: na1-1f.gif

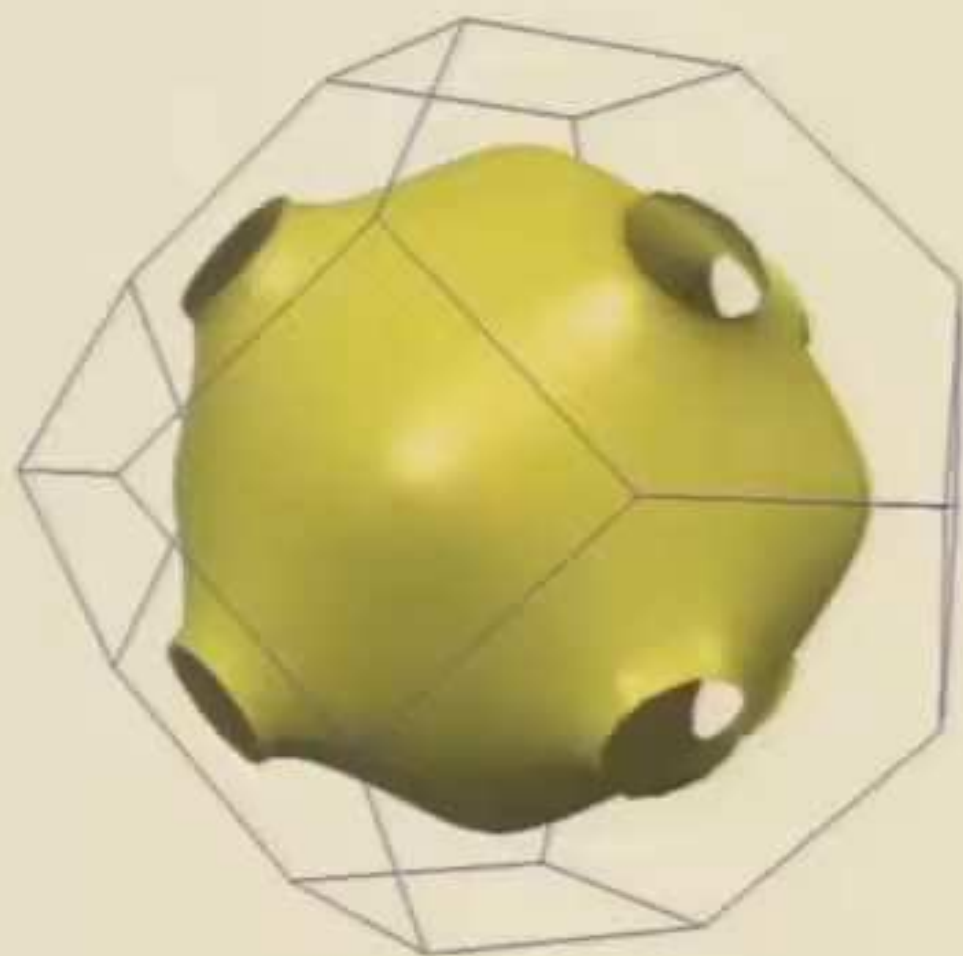


2d analogue

Fermi Surface of Sodium (BCC)

Monovalent = Half-Filled Brillouin Zone

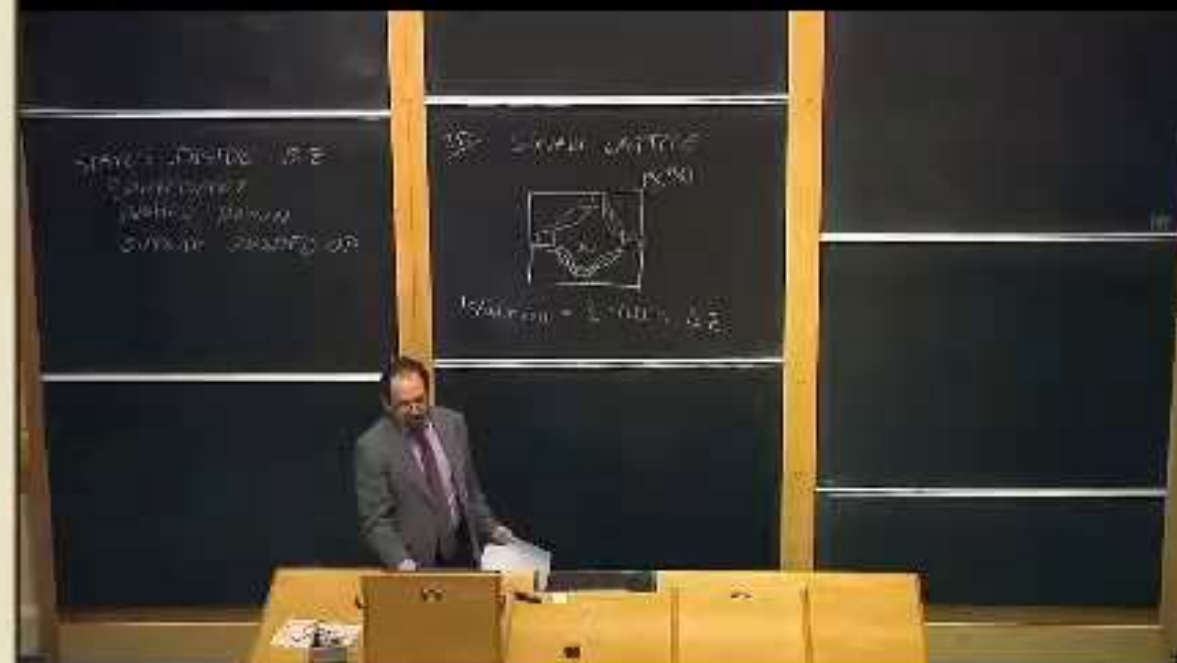




2d analogue

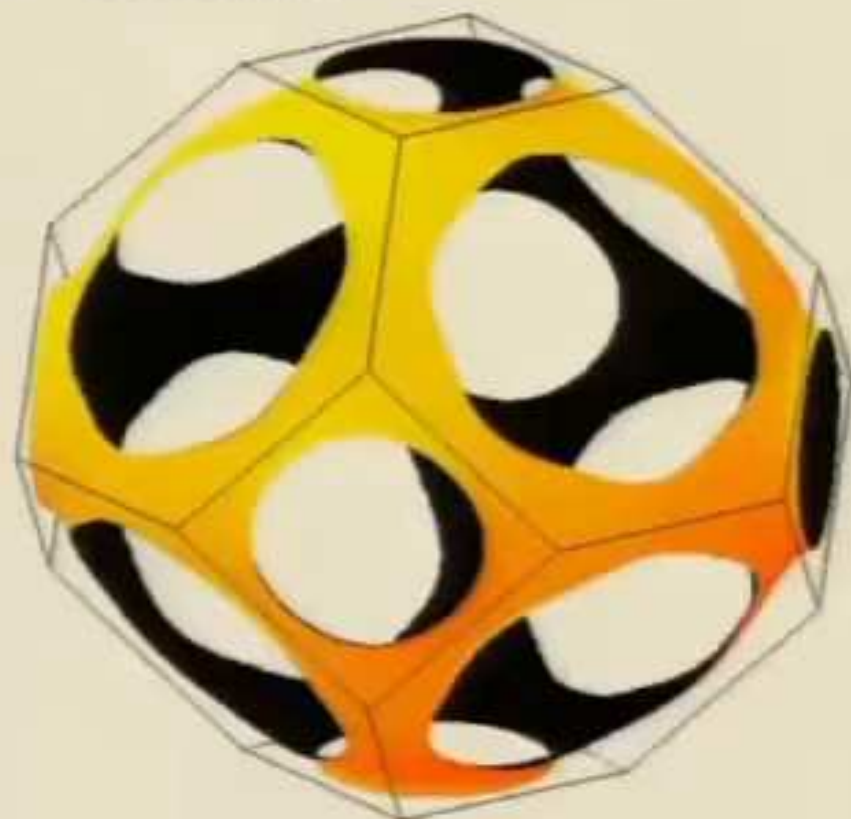
Fermi Surface of Copper (FCC)

Monovalent = Half-Filled Brillouin Zone



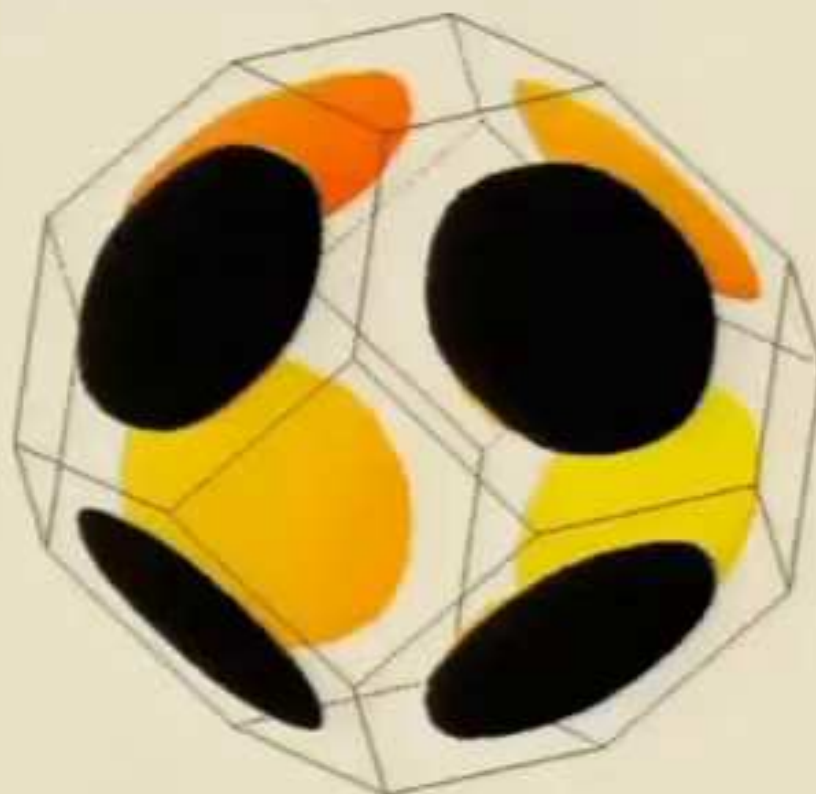


Ca
fcc: $a=10.53\text{au}$
1st band
electron Fermi surface
Fermi surface
file name: ca1-1f.gf



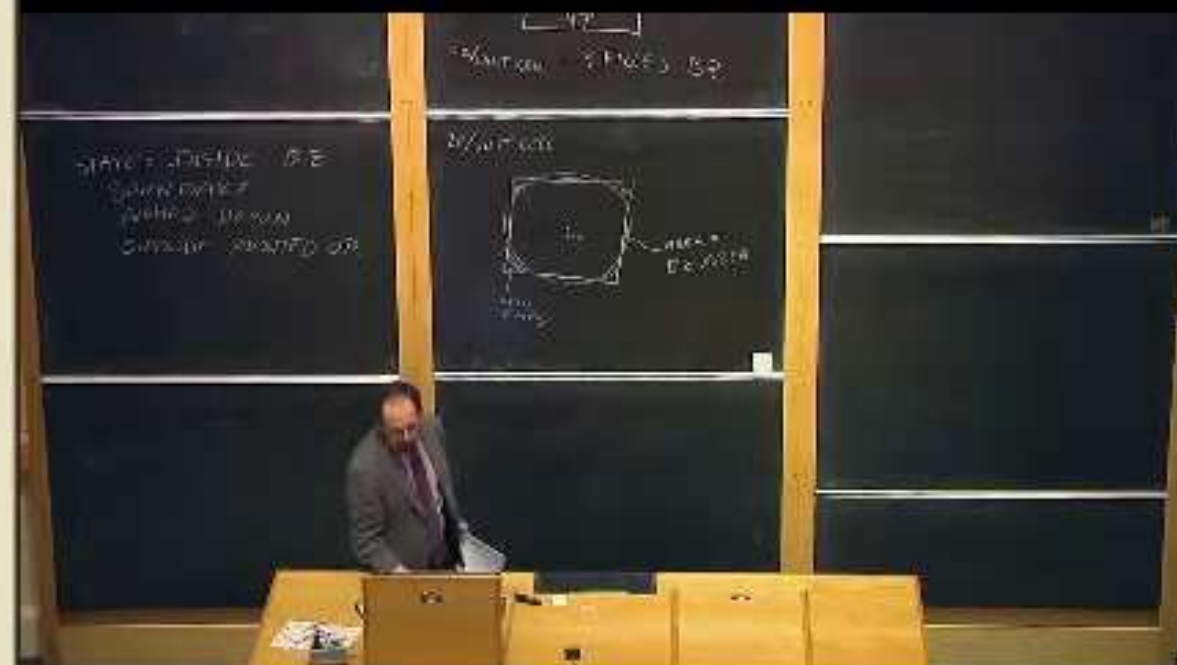
1st band

Ca
fcc: $a=10.53\text{au}$
2nd band
electron Fermi surface
Fermi surface
file name: ca1-2f.gf



2nd band

Calcium FCC
(Divalent)





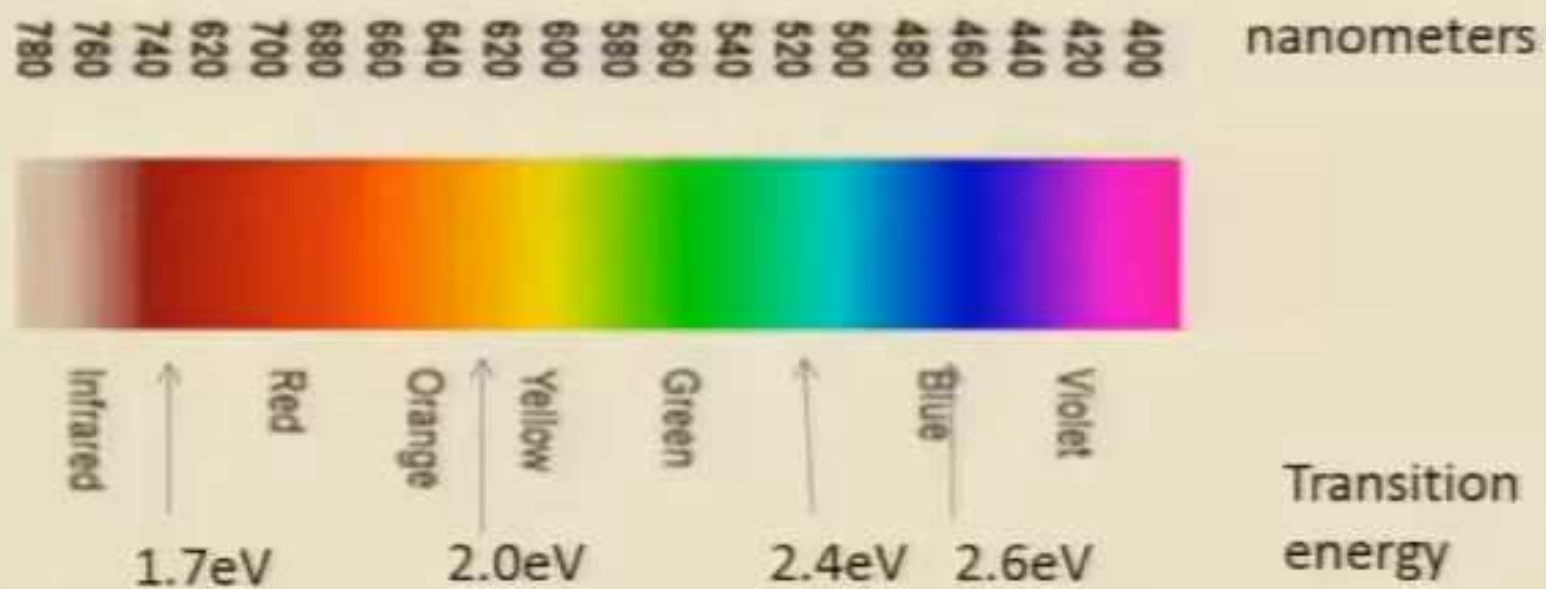
Cinnabar (HgS)
Gap = 2.0 eV

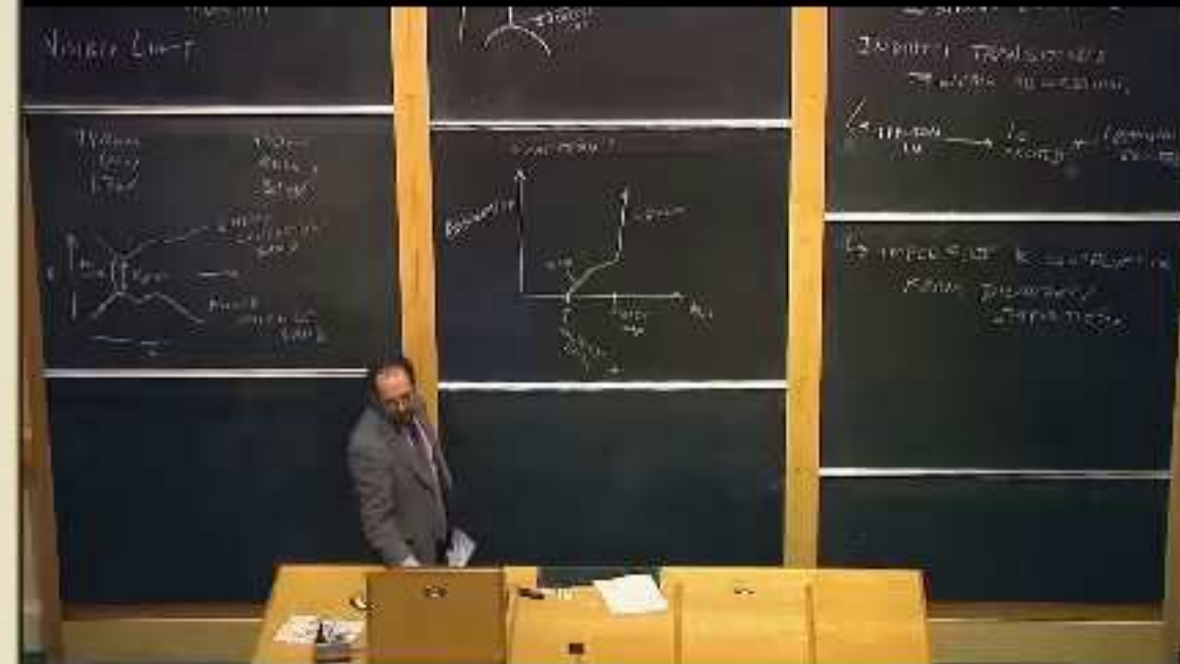
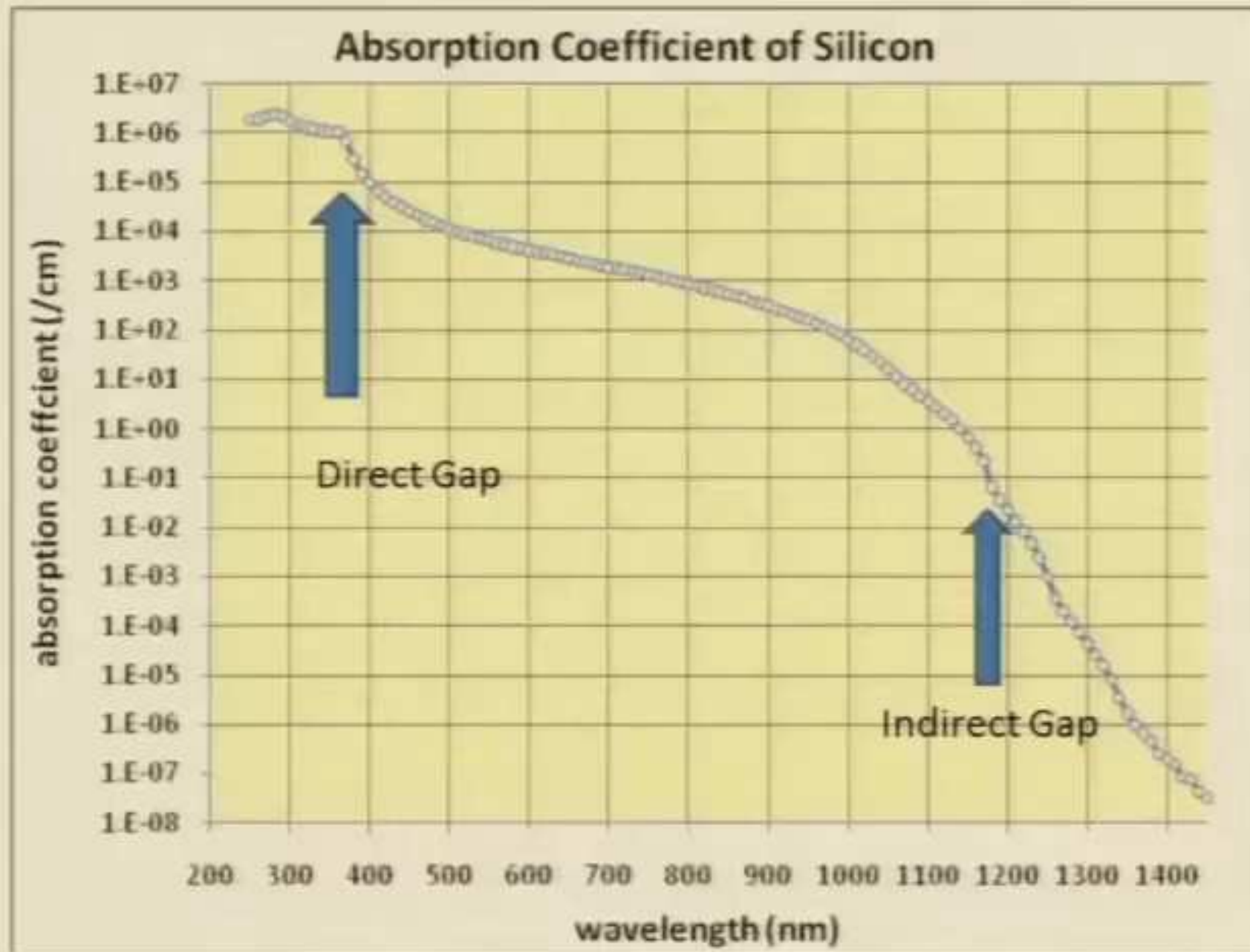


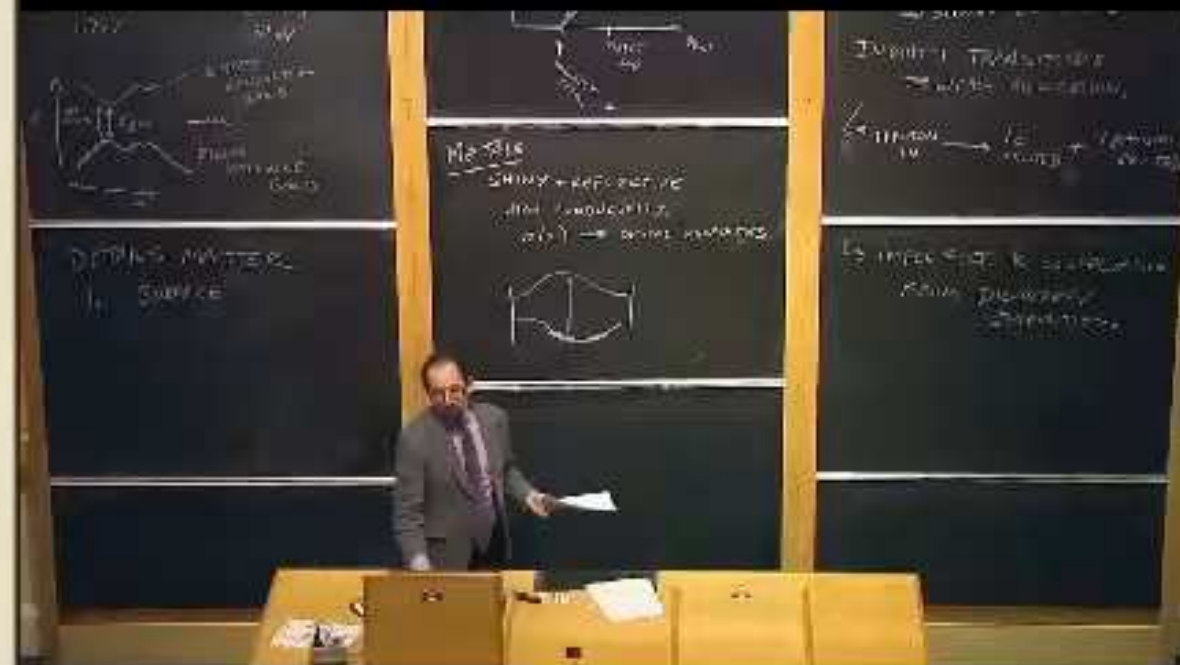
Realgar (As_4S_4)
Gap = 2.4 eV



Sulfur
Gap = 2.6 eV









Tiffany Diamond (Yellow)
Roughly 1 Nitrogen impurity per 10^6 carbon.

Estimated value 12 Million\$ (1983)



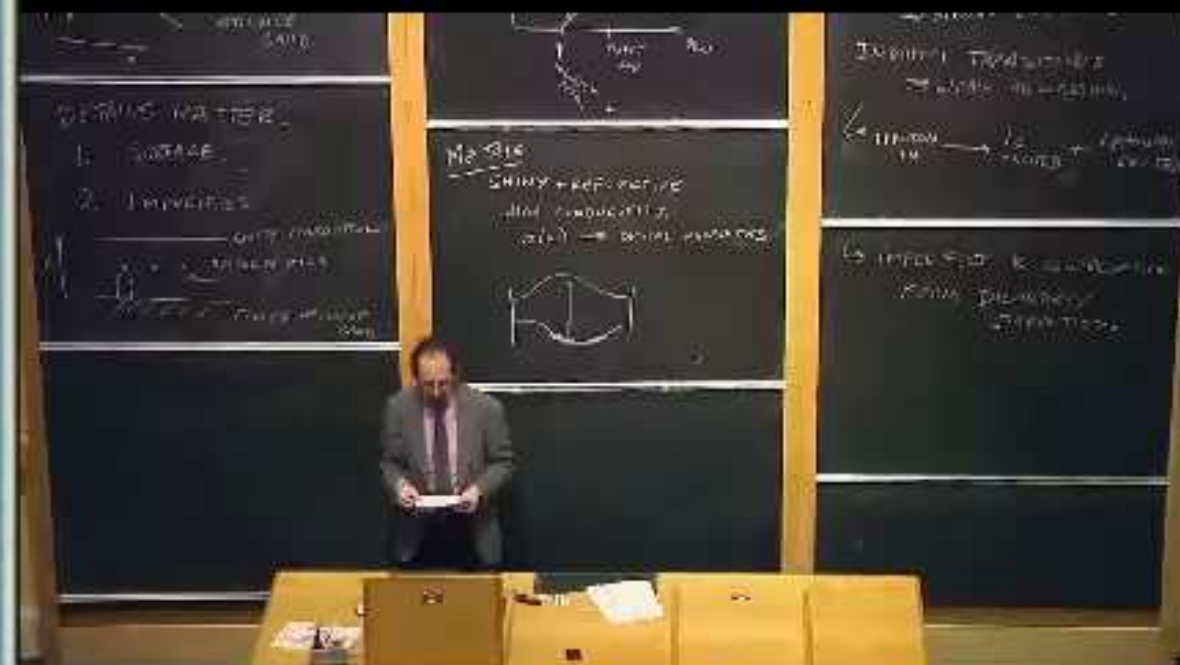
Hope Diamond (Blue)
Roughly 1 Boron impurity per 10^7 carbon.

Estimated current value = 250 Million \$



Synthetic Diamonds =

"Cheap" and any impurities
you want. (this example is
clear meaning no impurities)



Periodic Table of the Elements

Periodic Table of the Elements

1A																	0	
1	1 H																	2 He
2	3 Li	4 Be											5 B	6 C	7 N	8 O	9 F	10 Ne
3	11 Na	12 Mg	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar										
4	19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba	57 *La	72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr	88 Ra	89 +Ac	104 Rf	105 Ha	106 Sg	107 Ns	108 Hs	109 Mt	110 110	111 111	112 112	113 113					

* Lanthanide Series

+ Actinide Series

58 Ce	59 Pr	60 Nd	61 Pm	62 Sm	63 Eu	64 Gd	65 Tb	66 Dy	67 Ho	68 Er	69 Tm	70 Yb	71 Lu
90 Th	91 Pa	92 U	93 Np	94 Pu	95 Am	96 Cm	97 Bk	98 Cf	99 Es	100 Fm	101 Md	102 No	103 Lr

